Min Heap Sort

```
#include <iostream>

using namespace std;

// Function to maintain Min Heap

void heapify(int arr[], int n, int i)

{

        int smallest = i;

        int left = 2*i + 1;

        int right = 2*i + 2;

        if(left < n && arr[left] < arr[smallest])

                smallest = left;

        if(right < n && arr[right] < arr[smallest])

                smallest = right;

    if(smallest != i)

        {

                swap(arr[i], arr[smallest]);

                heapify(arr, n, smallest);

        }

}

// Min Heap Sort

void heapSort(int arr[], int n)

{

        // Build Min Heap

        for(int i = n/2 - 1; i >= 0; i--)

                heapify(arr, n, i);
```

```
	// Extract elements
	for(int i = n-1; i >= 0; i--)
	{
		swap(arr[0], arr[i]);
		heapify(arr, i, 0);
	}
}
// Print array
void printArray(int arr[], int n)
{
	for(int i = 0; i < n; i++)
		cout << arr[i] << " ";
}
int main()
{
	int arr[] = {12, 11, 13, 5, 6, 7};
	int n = sizeof(arr)/sizeof(arr[0]);
	heapSort(arr, n);
cout << "Sorted array: ";
	printArray(arr, n);
return 0;
}
```

Output

```
Sorted array: 13 12 11 7 6 5

=== Code Execution Successful ===
```